

LAPLACE AND FOURIER TRANSFORM

LAPLACE TRANSFORM

Laplace transform was named after a French mathematician, Pierre Simon Laplace. He and Samuel F. B. Morse first introduced Laplace transform and its application to differential equations.

The Laplace transform of a function $f(t)$ is denoted by $F(s)$ or $\mathcal{L}[f(t)]$.

$$\mathcal{L}[f(t)] = F(s) = \int_0^{\infty} f(t) e^{-st} dt \quad \text{--- (1)}$$

Where $f(t)$ = function in time domain.

$F(s) = \mathcal{L}[f(t)]$ = Laplace transform

$s = \sigma + j\omega$ = Complex Variable.

From the equation (1) we can say that the Laplace transform is an

"an integral transformation of $f(t)$ from the time domain into a complex frequency domain $F(s)$ ".

$$\mathcal{L}^{-1}[F(s)] = f(t) = \frac{1}{2\pi j} \int_{\sigma_1 - j\infty}^{\sigma_1 + j\infty} F(s) e^{st} ds \quad \text{--- (2)}$$

Equation 2 is the Inverse Laplace transform. To directly use this equation, the knowledge of Complex analysis is vital. Therefore the Laplace transform pair is

$$f(t) \longleftrightarrow F(s) \quad \text{--- (3)}$$

PROPERTIES OF LAPLACE TRANSFORM

1. Linearity

The linearity property is denoted by

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$$\mathcal{L}[a_1 f_1(t) + a_2 f_2(t)] = a_1 F_1(s) + a_2 F_2(s) \quad \text{--- (4)}$$

Where a_1 and a_2 = Constants.

$F_1(s)$ and $F_2(s)$ = Laplace transform of $f_1(t)$ and $f_2(t)$

(2) Scaling

The scaling property of Laplace transform is

$$\mathcal{L}[f(at)] = \int_0^{\infty} f(at) e^{-st} dt \quad \text{--- (5)}$$

where a = constant.

$$\text{If } x = at \quad \text{--- (6)}$$

$$\frac{dx}{dt} = a \quad \text{--- (7)}$$

$$\therefore dx = a dt \quad \text{--- (8)}$$

Substituting $x = at$ into eq 5

$$\mathcal{L}[f(at)] = \int_0^{\infty} f(x) e^{-x(s/a)} \frac{dx}{a} \quad \text{--- (9)}$$

$$\mathcal{L}[f(at)] = \frac{1}{a} \int_0^{\infty} f(x) e^{-x(s/a)} dx \quad \text{--- (10)}$$

$$\therefore \mathcal{L}[f(at)] = \frac{1}{a} F\left(\frac{s}{a}\right) \quad \text{--- (11)}$$

(3) Time shift

The time shift properties of Laplace transform is denoted by

$$\mathcal{L}[f(t-a)u(t-a)] = \int_0^{\infty} f(t-a)u(t-a) e^{-st} dt \quad \text{--- (12)}$$

but recall that for a unit function (shift) $u(t-a) = 0$ for $t < a$ and $u(t-a) = 1$ for $t > a$.

$$\mathcal{L}[f(t-a)u(t-a)] = \int_a^{\infty} f(t-a) e^{-st} dt \quad a > 0 \quad \text{--- (13)}$$

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Let $x = t - a \therefore t = x + a$.

Substituting in eq (13)

$$\mathcal{L}[f(t-a)u(t-a)] = \int_0^{\infty} f(x) e^{-s(x+a)} dx \quad (14)$$

$$\begin{aligned} \mathcal{L}[f(t-a)u(t-a)] &= e^{-as} \int_0^{\infty} f(x) e^{-sx} dx \\ &= e^{-as} F(s) \end{aligned} \quad (15)$$

④ Frequency Shift.

The frequency shift of Laplace transform is denoted by

$$\begin{aligned} \mathcal{L}[e^{-at} f(t) u(t)] &= \int_0^{\infty} e^{-at} f(t) e^{-st} dt \quad (16) \\ &= \int_0^{\infty} f(t) e^{-(s+a)t} dt \end{aligned}$$

$$\therefore \mathcal{L}[e^{-at} f(t) u(t)] = F(s+a) \quad (17)$$

⑤ Time differentiation.

The time differentiation property of Laplace transform is expressed

$$\mathcal{L}\left[\frac{df}{dt} u(t)\right] = \int_0^{\infty} \frac{df}{dt} e^{-st} dt \quad (18)$$

$$\text{Let } u = e^{-st}$$

$$\therefore \frac{du}{dt} = -s e^{-st} \quad (19)$$

$$\therefore du = -s e^{-st} dt \quad (20)$$

$$\text{Let } dv = \frac{df}{dt} dt \quad (21)$$

$$dv = df(t)$$

$$\therefore v = f(t) \quad (22)$$

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$$\begin{aligned} \mathcal{L}\left[\frac{df}{dt} u(t)\right] &= f(t) e^{-st} \Big|_0^{\infty} - \int_0^{\infty} f(t) [-s e^{-st}] dt \\ &= 0 - f(0^-) + s \int_0^{\infty} f(t) e^{-st} dt \end{aligned}$$

$$\therefore \mathcal{L}\left[\frac{df}{dt} u(t)\right] = s F(s) - f(0^-)$$

$$\mathcal{L}[f'(t)] = s F(s) - f(0^-) \quad \text{--- (23)}$$

The Laplace transform of second derivative

$$\mathcal{L}\left[\frac{\partial^2 f}{\partial t^2}\right] = s \mathcal{L}[f'(t)] - f'(0^-)$$

$$= s [s F(s) - f(0^-)] - f'(0^-)$$

$$\therefore \mathcal{L}[f''(t)] = s^2 F(s) - s f(0^-) - f'(0^-) \quad \text{--- (24)}$$

Therefore the Laplace transform of n th derivatives is given as

$$\mathcal{L}\left[\frac{\partial^n f}{\partial t^n}\right] = s^n F(s) - s^{n-1} f(0^-) - s^{n-2} f'(0^-) - \dots - s^0 f^{(n-1)}(0^-) \quad \text{--- (25) No}$$

⑥ Frequency differentiation

The frequency differentiation which is a property of Laplace transform can be derived as

$$F(s) = \int_0^{\infty} f(t) e^{-st} dt$$

$$\begin{aligned} \frac{\partial F(s)}{\partial s} &= \int_0^{\infty} f(t) (-t e^{-st}) dt \quad \text{--- (26)} \\ &= \int_0^{\infty} (-t f(t)) e^{-st} dt \end{aligned}$$

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f''
f'

$$= \mathcal{L}[-t f(t)]$$

$$\therefore \mathcal{L}[t f(t)] = - \frac{dF(s)}{ds} \quad \text{--- (27)}$$

(7) Time Periodicity.

If a signal $f(t)$ is periodic, it can be represented as the sum of time shifted functions.

$$\begin{aligned} f(t) &= f_1(t) + f_2(t) + f_3(t) + \dots \\ &= f_1(t) + f_1(t-T)u(t-T) + \\ &\quad f_1(t-2T)u(t-2T) + \dots \end{aligned} \quad \text{--- (28)}$$

The Laplace transform of $f(t)$ equals.

$$\begin{aligned} F(s) &= F(s) + F_1(s)e^{-Ts} + F_1(s)e^{-2Ts} + \dots \\ &= F_1(s) [1 + e^{-Ts} + e^{-2Ts} + \dots] \end{aligned} \quad \text{--- (29)}$$

$$\therefore F(s) = \frac{F_1(s)}{1 - e^{-Ts}} \quad \text{--- (30)}$$

Note $\left[1 + x + x^2 + x^3 + \dots = \frac{1}{1-x} \right]$

Geometric Series expansion.

(8) Initial and final values.

The Initial value of Laplace transform is

$$F(0^+) = \lim_{s \rightarrow \infty} s F(s) \quad \text{--- (31)}$$

The final value of Laplace transform is

$$F(\infty) = \lim_{s \rightarrow 0} s F(s) \quad \text{--- (32)}$$

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Table 1 Laplace transform properties

Property	$f(t)$	$F(s)$
Linearity	$a_1 f_1(t) + a_2 f_2(t)$	$a_1 F_1(s) + a_2 F_2(s)$
Time shift	$f(t-a)u(t-a)$	$e^{-as}F(s)$
Scaling	$f(at)$	$\frac{1}{a}F\left(\frac{s}{a}\right)$
Frequency shift	$e^{-at}f(t)$	$F(s+a)$
Time differentiation	$\frac{df}{dt}$	$sF(s) - f(0^-)$
	$\frac{\partial^2 f}{\partial t^2}$	$s^2 F(s) - sf(0^-) - f'(0^-)$
	$\frac{\partial^n f}{\partial t^n}$	$s^n F(s) - S^{n-1}f(0^-) - S^{n-2}f'(0^-) - \dots - f^{(n-1)}(0^-)$
Frequency differentiation	$t f(t)$	$\frac{d}{ds} F(s)$
Time periodicity	$f(t) = f(t+nT)$	$\frac{F_1(s)}{1 - e^{-sT}}$
Initial Value	$F(0^+)$	$\lim_{s \rightarrow \infty} s F(s)$
Final Value	$F(\infty)$	$\lim_{s \rightarrow 0} s F(s)$

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Worked examples. A

① Determine the Laplace transform of

(a) $u(t)$.

$$f(t) = u(t)$$

$$\mathcal{L}[u(t)] = \int_0^{\infty} 1 e^{-st} dt$$

$$= \left. \frac{-1}{s} e^{-st} \right|_0^{\infty}$$

$$= \frac{-1}{s}(0) - \left(\frac{-1}{s}(1) \right) \quad \text{Since } \lim_{t \rightarrow \infty} e^{-st} = 0 \text{ and } e^0 = 1$$

$$F(s) = \frac{1}{s}$$

② $e^{-at} u(t)$ at $a > 0$.

$$f(t) = e^{-at} u(t)$$

$$\mathcal{L}[e^{-at} u(t)] = \int_0^{\infty} e^{-at} e^{-st} dt$$

$$= \int_0^{\infty} e^{-(s+a)t} dt$$

$$= \left. \frac{-1}{s+a} e^{-(s+a)t} \right|_0^{\infty}$$

$$= \frac{-1}{s+a}(0) - \left(\frac{-1}{s+a}(1) \right)$$

$$F(s) = \frac{1}{s+a} \quad \therefore \mathcal{L}[e^{at} u(t)] = \frac{1}{s-a} \quad \text{Try this !!!}$$

③ $\delta(t)$

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$$f(t) = \delta(t)$$

$$\mathcal{L}[\delta(t)] = \int_{0^-}^{\infty} \delta(t) e^{-st} dt$$

$$= e^{-s \cdot 0} = 1 \quad \text{Since the } \delta = 0 \text{ everywhere else except at } t=0$$

$$(4) \quad r(t) = tu(t)$$

$$\mathcal{L}[r(t)] = \int_{0^-}^{\infty} t e^{-st} dt$$

$$\text{let } u = t$$

$$dv = e^{-st} dt$$

$$u = t \rightarrow du = dt$$

$$dv = e^{-st} dt \quad \therefore v = \frac{1}{s} e^{-st}$$

Using the Integration by part formular

$$= uv \Big|_0^{\infty} - \int_0^{\infty} v du$$

$$= \frac{-1}{s} t e^{-st} \Big|_0^{\infty} + \frac{1}{s} \int_0^{\infty} e^{-st} dt$$

$$= \frac{1}{s} \left[-\frac{1}{s} e^{-st} \right]_0^{\infty}$$

$$= \frac{1}{s} \left[0 + \frac{1}{s} \right]$$

$$= 1/s^2$$

$$\therefore \mathcal{L}(r(t)) = 1/s^2$$

$$(2) \quad f(t) = \sin wt \, u(t)$$

$$F(s) = \mathcal{L}[\sin wt] = \int_0^{\infty} (\sin wt) e^{-st} dt \quad \text{--- (33)}$$

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Using Euler's formula.

$$\sin \theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

$$\therefore \sin \omega t = \frac{e^{j\omega t} - e^{-j\omega t}}{2j}$$

Substituting into eq 33.

$$\mathcal{L}[\sin \omega t] = \int_0^{\infty} \left(\frac{e^{j\omega t} - e^{-j\omega t}}{2j} \right) e^{-st} dt$$

$$= \frac{1}{2j} \int_0^{\infty} (e^{j\omega t} e^{-st} - e^{-j\omega t} e^{-st}) dt$$

$$= \frac{1}{2j} \left[\frac{e^{-(s-j\omega)t}}{-(s-j\omega)} - \frac{e^{-(s+j\omega)t}}{-(s+j\omega)} \right]_0^{\infty}$$

$$= \frac{1}{2j} \left(\frac{1}{s-j\omega} - \frac{1}{s+j\omega} \right)$$

$$= \frac{1}{2j} \left(\frac{(s+j\omega) - (s-j\omega)}{(s-j\omega)(s+j\omega)} \right)$$

$$= \frac{1}{2j} \left(\frac{s+j\omega - s+j\omega}{s^2 - (j\omega)^2} \right)$$

$$= \frac{1}{2j} \left(\frac{2j\omega}{s^2 + \omega^2} \right)$$

$$\therefore \mathcal{L}[\sin \omega t] = F(s) = \frac{\omega}{s^2 + \omega^2}$$

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Assignment 1

Find the Laplace transform of
 $f(t) = \cos wt \, u(t)$.

(f) Find the Laplace transform of $f(t) = \delta(t) + 2u(t) - 3e^{-2t}$ at $t \geq 0$.

By linearity property.

$$F(s) = \mathcal{L}[\delta(t) + 2\mathcal{L}[u(t)] - 3\mathcal{L}[e^{-2t}u(t)]]$$

$$= 1 + 2 \times \frac{1}{s} - 3 \left(\frac{1}{s+2} \right)$$

$$= 1 + \frac{2}{s} - \frac{3}{s+2}$$

$$= \frac{s(s+2) + 2(s+2) - 3s}{s(s+2)}$$

$$= \frac{s^2 + 2s + 2s + 4 - 3s}{s(s+2)}$$

$$F(s) = \frac{s^2 + s + 4}{s(s+2)}$$

(g) Find the Laplace transform of $f(t) = \cos 2t + e^{-3t}$ at $t \geq 0$.

$$f(t) = \cos 2t + e^{-3t}$$

$$F(s) = \mathcal{L}[\cos 2t + e^{-3t}]$$

$$\text{Recall } \mathcal{L}[\cos wt] = \frac{s}{s^2 + w^2} \quad \text{and } \mathcal{L}[e^{-st}] = \frac{1}{s+a}$$

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$$F(s) = \frac{s}{s^2 + 2^2} + \frac{1}{s+3}$$

$$= \frac{s}{s^2+4} + \frac{1}{s+3}$$

$$= \frac{s(s+3) + s^2+4}{(s^2+4)(s+3)}$$

$$= \frac{s^2+3s+s^2+4}{(s^2+4)(s+3)}$$

$$\therefore F(s) = \frac{2s^2+3s+4}{(s+3)(s^2+4)}$$

Assignment 2.

Using frequency differentiation determine the laplace transform of

(a) $f(t) = t^2 \sin 2t u(t)$

(b) $f(t) = t^2 \cos 3t u(t)$

(h) Given fig 1, find the laplace transform of ~~the gate function~~ the gate function

from fig 1 it can be interpreted as

$$g(t) = 10 [u(t-2) - u(t-3)]$$

Using the time shift property

$$G(s) = 10 \left(\frac{e^{-2s}}{s} - \frac{e^{-3s}}{s} \right)$$

$$= \frac{10}{s} (e^{-2s} - e^{-3s})$$

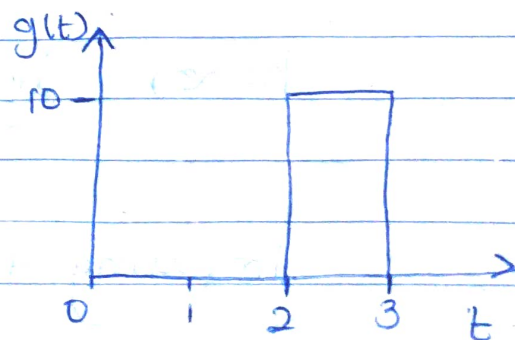


fig 1 gate function

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(i) Find the initial and final values of the function whose Laplace transform is

$$H(s) = \frac{20}{(s+3)(s^2+8s+25)}$$

For Initial Value theorem:

$$h(0) = \lim_{s \rightarrow \infty} sH(s) = \lim_{s \rightarrow \infty} \frac{20s}{(s+3)(s^2+8s+25)}$$

$$= \lim_{s \rightarrow \infty} \frac{20s}{s^3 + 11s^2 + 49s + 75}$$

divide the numerator and denominator by the highest power of s

$$= \lim_{s \rightarrow \infty} \frac{20s/s^3}{\frac{s^3}{s^3} + \frac{11s^2}{s^3} + \frac{49s}{s^3} + \frac{75}{s^3}}$$

Inputting the value $s = \infty$

$$= \frac{0}{1 + 0 + 0 + 0} = 0$$

Therefore the Initial value = 0

For final value

$$h(\infty) = \lim_{s \rightarrow 0} sH(s) = \lim_{s \rightarrow 0} \frac{20s}{(s+3)(s^2+8s+25)}$$

Inputting the value of $s = 0$

$$= \frac{0}{(0+3)(0+0+25)}$$

$$= 0$$

Therefore the final value is 0

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Inverse Laplace transform.

$$f(t) = \mathcal{L}^{-1}[F(s)]$$

(34)

Where $f(t)$ = function $\mathcal{L}^{-1}[F(s)]$ = Inverse laplace transform

Worked examples B.

a Find the Inverse laplace transform of

$$F(s) = \frac{3}{s} - \frac{5}{s+1} + \frac{6}{s^2+4}$$

$$\begin{aligned} f(t) &= \mathcal{L}^{-1}[F(s)] = \mathcal{L}^{-1}\left(\frac{3}{s}\right) - \mathcal{L}^{-1}\left(\frac{5}{s+1}\right) + \mathcal{L}^{-1}\left(\frac{6}{s^2+4}\right) \\ &= (3 - 5e^{-t} + 3\sin 2t)u(t) \quad t \geq 0 \end{aligned}$$

$$\textcircled{b} \quad F(s) = 1 + \frac{4}{s+3} - \frac{5s}{s^2+16}$$

$$\begin{aligned} f(t) &= \mathcal{L}^{-1}[F(s)] = \mathcal{L}^{-1}[1] + \mathcal{L}^{-1}\left(\frac{4}{s+3}\right) - \mathcal{L}^{-1}\left(\frac{5s}{s^2+16}\right) \\ &= \delta(t) + (4e^{-3t} - 5\cos 4t)u(t) \quad t \geq 0 \end{aligned}$$

$$\textcircled{c} \quad F(s) = \frac{-s^2+12}{s(s+2)(s+3)}$$

Firstly the partial fraction needs to be determined

$$\frac{s^2+12}{s(s+2)(s+3)} = \frac{A}{s} + \frac{B}{s+2} + \frac{C}{s+3}$$

Using Residue method

$$A = s F(s) \Big|_{s=0} = \frac{s^2+12}{(s+2)(s+3)} \Big|_{s=0} = \frac{0^2+12}{(0+2)(0+3)} = \frac{12}{6}$$

$$A = \underline{\underline{2}}$$

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$$B = \left. \frac{s+2}{s(s+3)} F(s) \right|_{s=-2} = \frac{s^2+12}{s(s+3)} \Big|_{s=-2} = \frac{4+12}{-2(-2+3)} = \frac{16}{-2} = -8$$

$$C = \left. \frac{s+3}{s(s+2)} F(s) \right|_{s=-3} = \frac{s^2+12}{s(s+2)} \Big|_{s=-3} = \frac{9+12}{-3(-3+2)} = \frac{21}{3} = 7$$

$$\text{Since } F(s) = \frac{A}{s} + \frac{B}{s+2} + \frac{C}{s+3}$$

Inputting the values of A, B and C

$$F(s) = \frac{2}{s} - \frac{8}{s+2} + \frac{7}{s+3}$$

$$\therefore f(t) = \mathcal{L}^{-1} F(s) = (2 - 8e^{-2t} + 7e^{-3t}) u(t) \quad t \geq 0$$

Try this !!!

$$\text{Find } f(t) \text{ for } F(s) = \frac{6(s+2)}{(s+1)(s+3)(s+4)}$$

$$\text{Find } g(t) \text{ for } G(s) = \frac{10s^2 + 4}{s(s+1)(s+2)^2}$$

APPLICATION OF LAPLACE TRANSFORM.

Laplace transform is one of the most powerful tool (mathematical) used. In analysis, synthesis and design. In this section we would be using knowledge in laplace transform and its inverse to analyze circuits. This can be achieved using three main steps

- ① The circuit is transformed from the time domain to frequency domain.
- ② The circuit is solved using any circuit analysis technique.
- ③ The Inverse transform is taken to obtain the solution in time domain.

One advantage of using Laplace transform in circuit analysis is that the initial conditions are part of the transformation. The complete response (transient and steady state) can also be obtained.

$$V(t) = R i(t) \quad \text{--- (35)}$$

$$V(s) = R I(s) \quad \text{--- (36)}$$

For a capacitor

$$V(t) = C \frac{dv(t)}{dt} \quad \text{--- (37)}$$

$$I(s) = C [s V(s) - V(0^-)] \quad \text{--- (38)}$$

$$= s C V(s) - C V(0^-)$$

$$\therefore V(s) = \frac{1}{sC} I(s) + \frac{V(0^-)}{s} \quad \text{--- (39)}$$

For an inductor.

$$V(t) = L \frac{di(t)}{dt} \quad \text{--- (40)}$$

$$V(s) = L [s I(s) - i(0^-)] \quad \text{--- (41)}$$

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$$I(s) = \frac{1}{sL} V(s) + \frac{i(0)}{s} \quad \text{--- (42)}$$

If we assume zero Initial Conditions

Resistor $V(s) = RI(s)$

Inductor $V(s) = sLI(s)$

Capacitor $V(s) = \frac{1}{sC} I(s)$

(43)

The Impedance in the s domain is defined as

Resistor $Z(s) = R$

Capacitor $Z(s) = 1/sC$

Inductor $Z(s) = sL$

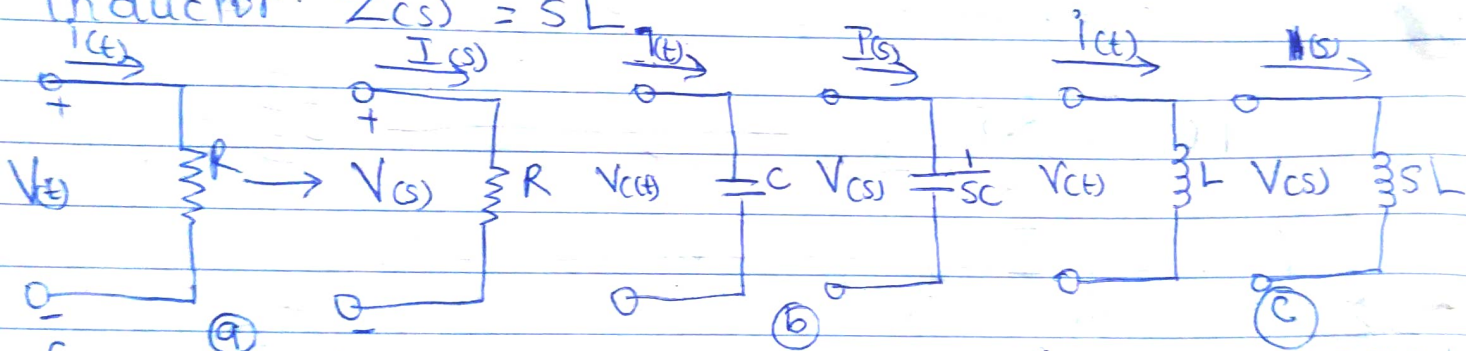


fig 2 a, b, and c. Time and s domain represents of passive elements under zero Initial Condition.

Worked Examples c

(a) Find $V_o(t)$ in fig 3 assuming zero Initial Condition

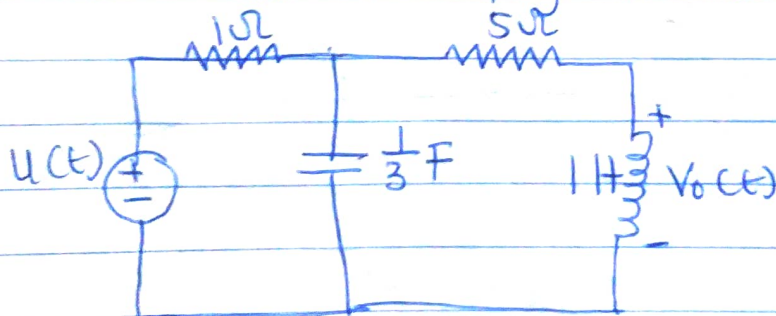


fig 3.

To solve this, The circuit needs to be transformed from the time domain to s domain

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$$u(t) = \frac{1}{s}$$

$$1 \text{ H} = sL = 1s = s$$

$$\frac{4}{3} \text{ F} = \frac{1}{sC} = \frac{1}{s \frac{1}{3}} = \frac{3}{s}$$

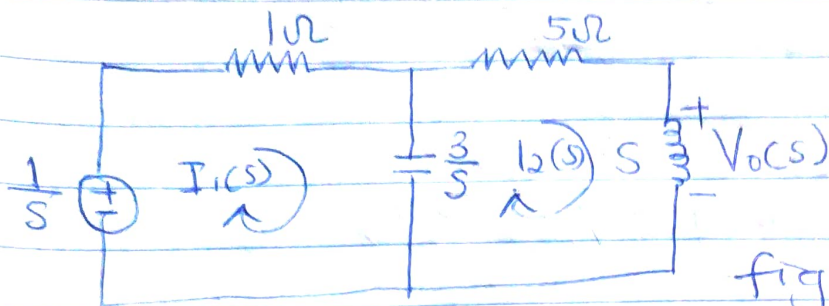


fig 4

Applying mesh analysis.

for mesh 1

$$\frac{1}{s} = \left(1 + \frac{3}{s}\right) I_1 - \frac{3}{s} I_2 \quad \text{--- (44)}$$

for mesh 2

$$0 = -\frac{3}{s} I_1 + \left(s + 5 + \frac{3}{s}\right) I_2$$

$$\frac{3}{s} I_1 = \left(s + 5 + \frac{3}{s}\right) I_2$$

$$I_1 = \frac{1}{3} (s^2 + 5s + 3) I_2 \quad \text{--- (45)}$$

Sub for I_1 into eq. (44)

$$\frac{1}{s} = \left(1 + \frac{3}{s}\right) \frac{1}{3} (s^2 + 5s + 3) I_2 - \frac{3}{s} I_2$$

To eliminate the denominator we multiply by $3s$

$$3 = (s^3 + 8s^2 + 18s) I_2 - 9 I_2$$

$$\therefore I_2 = \frac{3}{s^2 + 8s^2 + 18s}$$

$$V_o(s) = s I_2$$

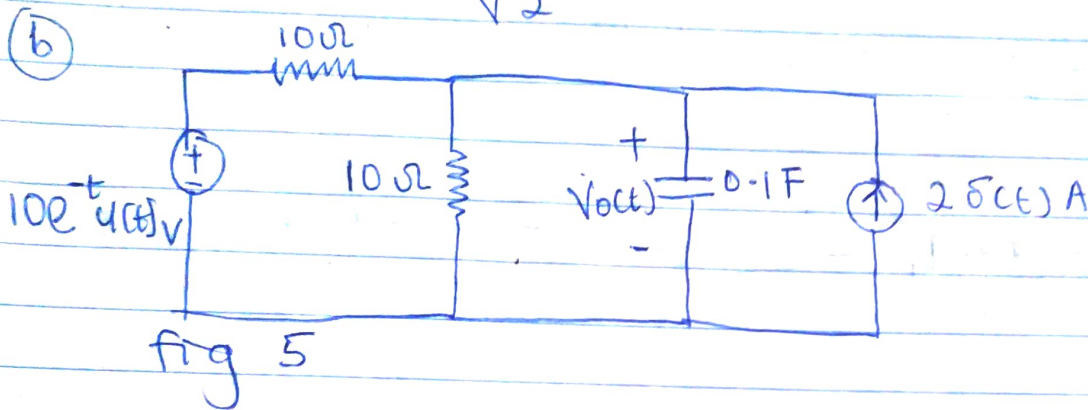
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$$= \frac{3}{s^2 + 8s + 18}$$

$$V_o(s) = \frac{3}{\sqrt{2}} \frac{1}{(s+4)^2 + (\sqrt{2})^2}$$

Taking Inverse transform

$$V_o(t) = \frac{3}{\sqrt{2}} e^{-4t} \sin \sqrt{2}t \text{ V } t \geq 0$$



Find $V_o(t)$ in the circuit of Fig 5 assuming $V_o(0) = 5V$.

First the circuit needs to be transformed into the s -domain.

$$10e^{-t} u(t) = \frac{10}{s+1}$$

$$10 \Omega = 10 \Omega$$

$$0.1 F = \frac{1}{sC} = \frac{1}{s \cdot 0.1} = \frac{10}{s}$$

$$V_o(0) = 5V$$

$C V_o(0) = 0.1(5) = 0.5A$ which will be in parallel with the capacitor impedance.

Using KCL at $V_o(s)$ considering current source representation

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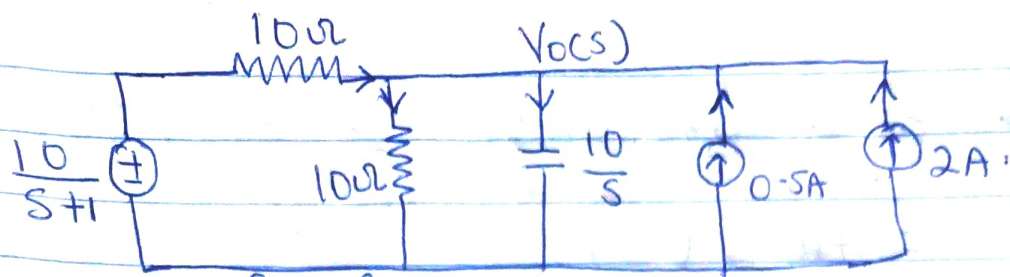


fig 6

$$\left(\frac{\frac{10-V_0}{s+1} - \frac{V_0}{10} - \frac{V_0}{10/s} + 2 + 0.5 \right) = 0$$

$$\frac{10-V_0}{s+1} + 2 + 0.5 = \frac{V_0}{10} + \frac{V_0}{10/s}$$

$$\frac{10}{s+1} - V_0 + 2.5 = \frac{V_0}{10} + \frac{sV_0}{10}$$

10. 25 + 25 =

Multiply by 10 to eliminate the denominator.

$$\frac{10}{s+1} - V_0 + 25 = V_0 + sV_0$$

$$\frac{10}{s+1} + 25 = V_0 + sV_0 + V_0$$

$$\frac{10}{s+1} + 25 = V_0(s+2)$$

$$V_0 = \frac{\frac{10}{s+1} + 25}{s+2}$$

$$V_0 = \frac{10 + 25s + 25}{(s+1)(s+2)}$$

$$= \frac{25s + 35}{(s+1)(s+2)}$$

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Simplifying using partial fraction.

$$V_0 = \frac{25s + 35}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2}$$

$$A = (s+1)V_0(s) \Big|_{s=-1} = \frac{25s + 35}{s+2} \Big|_{s=-1}$$

$$= \frac{-25 + 35}{-1 + 2} = \underline{10}$$

$$B = (s+2)V_0(s) \Big|_{s=-2} = \frac{25s + 35}{s+1}$$

$$= \frac{-50 + 35}{-2 + 1} = \underline{15}$$

$$V_0(s) = \frac{10}{s+1} + \frac{15}{s+2}$$

Taking the inverse Laplace.

$$\mathcal{L}^{-1}V_0(s) = V_0(t) = (10e^{-t} + 15e^{-2t})u(t) \text{ V.}$$

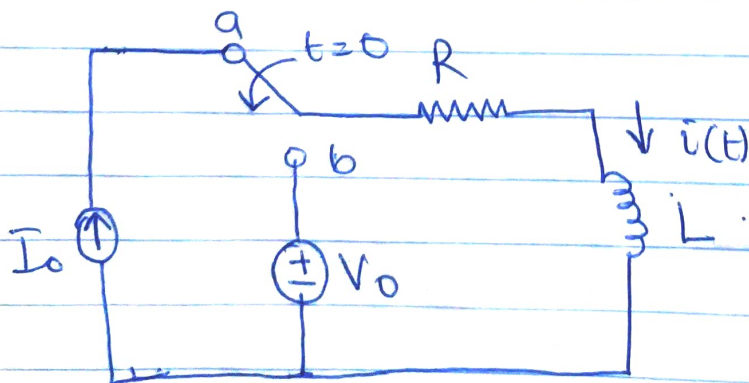
Practice problem.

fig 7

From fig 7 if the switch moves from position a

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to position b at $t=0$, Find $i(t)$ for $t > 0$.

Table 2. Laplace transform Pairs

	function	Laplace transform
1	$f(t)$	$F(s)$
2	$\delta(t)$	1
3	$u(t)$	$1/s$
4	e^{-at}	$1/(s+a)$
5	t	$1/s^2$
6	t^n	$\frac{n!}{s^{n+1}}$
7	$t e^{-at}$	$1/(s+a)^2$
8	$t^n e^{-at}$	$\frac{n!}{(s+a)^{n+1}}$
9	$\sin wt$	$\frac{w}{s^2 + w^2}$
10	$\cos wt$	$\frac{s}{s^2 + w^2}$
11	$\sin(wt + \phi)$	$\frac{s \sin \phi + w \cos \phi}{s^2 + w^2}$
12	$\cos(wt + \phi)$	$\frac{s \cos \phi - w \sin \phi}{s^2 + w^2}$
13	$e^{-at} \sin wt$	$\frac{w}{(s+a)^2 + w^2}$
14	$e^{-at} \cos wt$	$\frac{s+a}{(s+a)^2 + w^2}$

FOURIER TRANSFORM

Fourier allows the concept of frequency spectrum to non periodic signal. It is an integral representation of a non periodic function.

NB Revise Fourier Series.

from the exponential Fourier series

$$f(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t} \quad \text{--- (46)}$$

$$C_n = \frac{1}{T} \int_{-T/2}^{T/2} f(t) e^{-jn\omega_0 t} dt \quad \text{--- (47)}$$

but $\omega_0 = \frac{2\pi}{T}$

$\omega_0 = \text{angular frequency}$
 $T = \text{Period}$

Sub of 47 into 46

$$\begin{aligned} f(t) &= \sum_{n=-\infty}^{\infty} \left[\frac{1}{T} \int_{-T/2}^{T/2} f(t) e^{-jn\omega_0 t} dt \right] e^{jn\omega_0 t} \quad \text{--- (48)} \\ &= \sum_{n=-\infty}^{\infty} \left[\frac{\Delta\omega}{2\pi} \int_{-T/2}^{T/2} f(t) e^{-jn\omega_0 t} dt \right] e^{jn\omega_0 t} \end{aligned}$$

Where $\Delta\omega = (n+1)\omega_0 - n\omega_0 = \omega_0 = \frac{2\pi}{T}$
 $\therefore \Delta\omega = \frac{2\pi}{T}$

$$= \frac{1}{2\pi} \sum_{n=-\infty}^{\infty} \left[\int_{-T/2}^{T/2} f(t) e^{-jn\omega_0 t} dt \right] \Delta\omega e^{jn\omega_0 t}$$

as $T \rightarrow \infty$

$$\sum_{n=-\infty}^{\infty} \rightarrow \int_{-\infty}^{\infty}$$

$$\begin{aligned} \Delta\omega &\rightarrow d\omega \\ n\omega_0 &\rightarrow \omega \end{aligned}$$

--- (49)

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Sub of (49) into (48)

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt \right] e^{j\omega t} d\omega \quad (50)$$

$$F(\omega) = \mathcal{F}[f(t)] = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt \quad (51)$$

Where $F(\omega)$ = Fourier transform $\mathcal{F}$ = Fourier transform operator.The Inverse Fourier transform $\mathcal{F}^{-1}$.

$$f(t) = \mathcal{F}^{-1}[F(\omega)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega \quad (52)$$

$$f(t) \longleftrightarrow F(\omega)$$

Worked examples - D

Find the Fourier transform of the following function.

(a) $\delta(t - t_0)$

This is an impulse function.

$$F(\omega) = \mathcal{F}[\delta(t - t_0)] = \int_{-\infty}^{\infty} \delta(t - t_0) e^{-j\omega t} dt$$

applying shifting theorem

$$\int_{-\infty}^{\infty} f(t) \delta(t - t_0) dt = f(t_0) \quad F(\omega) = e^{-j\omega t_0}$$

Assuming $t_0 = 0$

$$\mathcal{F}[\delta(t)] = 1$$

(b) $e^{j\omega_0 t}$

$$\int_{-\infty}^{\infty} e^{j\omega_0 t} d\omega = 2\pi \delta(t)$$

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#/III Interchanging t and w

$$\int_{-\infty}^{\infty} e^{j\omega t} dt = 2\pi \delta(\omega)$$

$$F[e^{j\omega_0 t}] = \int_{-\infty}^{\infty} e^{j\omega_0 t} e^{-j\omega t} dt$$

$$= \int_{-\infty}^{\infty} e^{j(\omega_0 - \omega)t} dt = 2\pi \delta(\omega_0 - \omega)$$

The impulse function is an even function with $\delta(\omega_0 - \omega) = \delta(\omega - \omega_0)$.

$$F[e^{j\omega_0 t}] = 2\pi \delta(\omega - \omega_0)$$

Making $\omega_0 = -\omega_0$

$$F[e^{-j\omega_0 t}] = 2\pi \delta(\omega + \omega_0)$$

Let $\omega_0 = 0$

$$F[1] = 2\pi \delta(\omega)$$

© $\cos \omega_0 t$

To solve this we introduce Euler's formula

$$\cos(\theta) = \frac{e^{j\theta} + e^{-j\theta}}{2} \quad \text{--- (53)}$$

Sub θ as $\omega_0 t$

$$\cos(\omega_0 t) = \frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2}$$

Using linearity property

$$F[af(t) + bg(t)] = aF[f(t)] + bF[g(t)]$$

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$$\begin{matrix} \infty & & \infty \\ & \searrow & \nearrow \\ & 0 & \\ & \nearrow & \searrow \\ -\infty & & -\infty \end{matrix}$$

$$F[\cos(\omega_0 t)] = F\left[\frac{1}{2} e^{j\omega_0 t} + \frac{1}{2} e^{-j\omega_0 t}\right]$$

$$= \frac{1}{2} F[e^{j\omega_0 t}] + \frac{1}{2} F[e^{-j\omega_0 t}] \quad \text{--- (54)}$$

but we know that

$$F[e^{j\omega_0 t}] = 2\pi \delta(\omega - \omega_0)$$

(55)

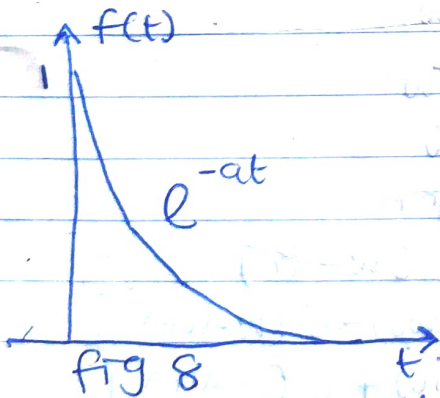
$$\therefore F[e^{-j\omega_0 t}] = 2\pi \delta(\omega + \omega_0)$$

Sub eq (55) into (54)

$$F[\cos(\omega_0 t)] = \frac{1}{2} [2\pi \delta(\omega - \omega_0)] + \frac{1}{2} [2\pi \delta(\omega + \omega_0)]$$

$$\therefore F[\cos(\omega_0 t)] = \pi \delta(\omega - \omega_0) + \pi \delta(\omega + \omega_0) \quad \text{--- (56)}$$

(d) Obtain the Fourier transform of a "switched on" exponential function from fig 8



$$f(t) = e^{-at} u(t)$$

This can be represented functionally as

$$f(t) = e^{-at} u(t) = \begin{cases} e^{-at} & t \geq 0 \\ 0 & t < 0 \end{cases}$$

$$F[f(t)] = F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

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$$F(\omega) = \int_0^{\infty} e^{-at} e^{-j\omega t} dt$$

$$= \int_0^{\infty} e^{-(a+j\omega)t} dt$$

$$= \frac{-1}{a+j\omega} e^{-(a+j\omega)t} \Big|_0^{\infty}$$

(57)

$$\therefore F(\omega) = \frac{1}{a+j\omega}$$

(58)

Table 3 Fourier Transform Pairs

S/No	Function $f(t)$	Fourier transform $F(\omega)$
1	$\delta(t)$	1
2	1	$2\pi\delta(\omega)$
3	$u(t)$	$\pi\delta(\omega) + \frac{1}{j\omega}$
4	$u(t+\tau) - u(t-\tau)$	$2 \frac{\sin \omega\tau}{\omega}$
5	t	$-2/\omega^2$
6	$\text{sgn}(t)$	$2/j\omega$
7	$e^{-at} u(t)$	$\frac{1}{a+j\omega}$
8	$e^{at} u(-t)$	$\frac{1}{a-j\omega}$
9	$t^n e^{-at} u(t)$	$\frac{n!}{(a+j\omega)^{n+1}}$
10	$e^{j\omega_0 t}$	$2\pi\delta(\omega - \omega_0)$
11	$\sin \omega_0 t$	$j\pi[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$
12	$\cos \omega_0 t$	$\pi[\delta(\omega + \omega_0) + \delta(\omega - \omega_0)]$
13	$e^{-at} \sin \omega_0 t u(t)$	$\frac{\omega_0}{(a+j\omega)^2 + \omega_0^2}$
14	$e^{-at} \cos \omega_0 t u(t)$	$\frac{a+j\omega}{(a+j\omega)^2 + \omega_0^2}$

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Assignment 3.

- ① Discuss the properties of fourier transform
- ② Compare Fourier transform and laplace transform.

APPLICATION OF FOURIER TRANSFORM IN CIRCUIT ANALYSIS.

Fourier transform in circuit analysis can be obtained by transforming the circuit element into the frequency domain and take the fourier transform of the excitation.

$$R \rightarrow R$$

$$L \Rightarrow j\omega L$$

$$C \rightarrow \frac{1}{j\omega C}$$

(59)

Fourier transform produces a response for $-\infty < t < \infty$, but on the other hand it cannot handle circuit with initial condition. To solve this the transfer function comes into play.

The transfer function

$$H(\omega) = \frac{Y(\omega)}{X(\omega)}$$

(60)

Where $H(\omega)$ = transfer function.

$Y(\omega)$ = Output response.

$X(\omega)$ = Input excitation.

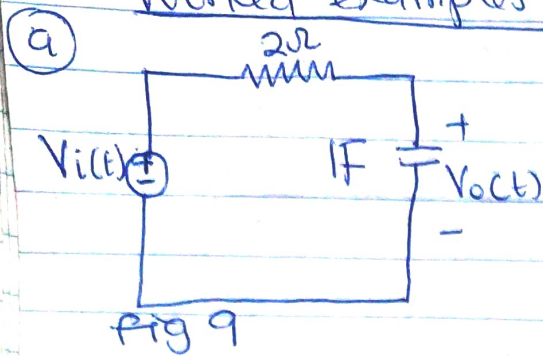
$$Y(\omega) = H(\omega) X(\omega)$$

(61)

$$Y(\omega) = H(\omega) = \mathcal{F}[h(t)]$$

(62)

Eq 62 shows that $H(\omega)$ is the fourier transform of the impulse response $h(t)$.

Worked examples · E

Find $V_o(t)$ in the circuit in Fig 9. For $V_i(t) = 2e^{-3t} u(t)$

Fourier transform of the input voltage

$$V_i(\omega) = \frac{2}{3 + j\omega} \quad * \text{ from the table}$$

The transfer function $H(\omega) = \frac{V_o(\omega)}{V_i(\omega)}$

The circuit consists of a 2Ω resistor in series with a $1F$ capacitor.

Impedance $R = 2\Omega$

$$Z_c = \frac{1}{j\omega C} \quad \text{but } C = 1$$

$$= \frac{1}{j\omega}$$

$$H(\omega) = \frac{V_o(\omega)}{V_i(\omega)}$$

$$= \frac{Z_c}{R + Z_c} = \frac{1/j\omega}{2 + 1/j\omega}$$

$$* \quad = \frac{1}{2j\omega + 1}$$

$$V_o(\omega) = H(\omega) V_i(\omega)$$

$$= \left(\frac{1}{1 + 2j\omega} \right) \left(\frac{2}{3 + j\omega} \right)$$

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$$= \frac{2}{(1+2j\omega)(3+j\omega)}$$

Using partial fraction expansion to solve

$$\frac{2}{(1+2j\omega)(3+j\omega)} = \frac{A}{1+2j\omega} + \frac{B}{3+j\omega}$$

(63)

solving for A and B

$$2 = A(3+j\omega) + B(1+2j\omega)$$

let $j\omega = -3$ to eliminate the A part

$$2 = A(3-3) + B(1+2(-3))$$

$$2 = B(-5)$$

$$B = \frac{-2}{5}$$

let $j\omega = -1/2$ to eliminate the B part.

$$2 = A(3 - 1/2)$$

$$2 = A(5/2)$$

$$\therefore A = \frac{4}{5}$$

Sub A and B into eq 63

$$V_o(\omega) = \frac{\frac{4}{5}}{1+2j\omega} - \frac{\frac{2}{5}}{3+j\omega}$$

$$= \frac{\frac{2}{5}}{1/2 + j\omega} - \frac{\frac{2}{5}}{3+j\omega}$$

Finding the Inverse Fourier transform properties,

$$\mathcal{F}^{-1} \left\{ \frac{1}{a+j\omega} \right\} = e^{-at} u(t)$$

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{Saturday}

$$\text{Then } V_0 = \mathcal{F}^{-1} V_0(\omega)$$

$$= \frac{2}{5} e^{-t/2} u(t) - \frac{2}{5} e^{-3t} u(t)$$

$$V_0(t) = \frac{2}{5} (e^{-t/2} - e^{-3t}) u(t).$$